



ACT 3 • THE STACK

# The Supply Chain of Intelligence.

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10 layers • 50 sublayers • 4 structural laws.

Below the line is foundation. Above is where intelligence compounds.

A LEARNING DECK FOR PRODUCT LEADERS, FOUNDERS & INVESTORS  
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# A common language for the generative AI stack.

Today the industry speaks in shorthand — 'wrapper', 'agent', 'copilot' — that hides where value actually accrues. This deck names every layer the same way every time, so product teams, founders and investors can talk about the same thing without translation.

One page per layer. Same color appears in every diagram on [supplychainofai.com](https://supplychainofai.com). Sublayers marked with ★ are the 18 structurally defensible positions — the moat map.

Three jobs the deck does:

## FOR PRODUCT LEADERS

Map your current product to layers. Highlight where you depend on a layer you don't own. Decide what to build, what to rent, what to abandon.

## FOR FOUNDERS

Place your wedge on exactly one or two layers. Show the path up the stack. Investors who know this framework can pattern-match in 30 seconds.

## FOR INVESTORS

Score companies on layer position × Law II (bottleneck) × Law IV (separation). Three layers = sharper diligence than 30 KPIs.

L-1

Resources

The Mines.

- L-1a

Energy

Power generation and procurement for AI compute
- L-1b

Thermal & Water Management

Cooling, water access, heat dissipation at scale
- L-1c

Fabrication & Foundry

Chip fab capacity and semiconductor manufacturing
- L-1d

Critical Materials

Rare earths, lithium, cobalt, specialized substrates
- L-1e

Skilled Trades & Human Capital

Electricians, technicians, data-center builders

VERDICT

The real bottleneck. Slow to build, impossible to fake.

HOW PRODUCT LEADERS USE THIS LAYER

- Ask: which resource constraint actually gates my product (power, fab, talent)?
- If you're a buyer, treat L-1 as macro risk, not roadmap.
- If you're an investor, time horizons here are decades.

L0

# Infrastructure

The Shovels.

L0a

## Silicon

GPUs, TPUs, custom AI accelerators

L0b

## Data Centers

Physical facilities housing compute at scale

L0c

## Interconnect Fabric

Networking between chips, racks, regions

L0d

## Compute Access Cloud

On-demand compute rental and scheduling

L0e

## Edge & On-Device

Local inference on phones, vehicles, sensors

### VERDICT

Shovel sellers win every gold rush.

### HOW PRODUCT LEADERS USE THIS LAYER

- Treat as 'rent, don't own' for >99% of teams.
- Watch for hyperscaler bundle creep — it eats L2/L5/L6 over time.
- If your unit economics depend on inference cost, lock multi-cloud optionality.

L1

# Data

The Ore.

L1a

## Public & Open Data

Common Crawl, Wikipedia, gov data

L1b

## Proprietary Data

Licensed, paywalled, internally generated corpora

L1c

## Behavioral Data

Clicks, sessions, usage patterns, interaction logs

L1d

## Outcome Data

Labels, results, conversions — what actually happened

L1e

## Synthetic Data

Machine-generated data for training & simulation

⚠ = DEFENSIBLE SUBLAYER (PART OF THE 18-POSITION MOAT MAP)

### VERDICT

Structurally safe. API-first wins.

### HOW PRODUCT LEADERS USE THIS LAYER

- The fastest moat test: would a top-3 lab pay you for your data?
- Public data is not a moat. Behavioral + outcome data is.
- Instrument now — proprietary data only compounds with time.

## L2

# Models

## The Smelter.

**L2a Foundation Models**

GPT, Claude, Gemini, Llama

**L2b Specialized Models**

Domain-tuned or distilled vertical models

**L2c Embedding & Retrieval**

Vector reps, search indices, RAG infra

**L2d Model Routing & Composition**

Selecting, chaining, ensembling models

**L2e Reasoning Models**

Extended chain-of-thought, planning, multi-step

## VERDICT

Winner-take-most. Commodity risk high.

## HOW PRODUCT LEADERS USE THIS LAYER

- Default to renting models. Build only when scale economics + quality demand it.
- Assume the model below you will absorb your wrapper. Plan the next layer.
- Track release cycles — your differentiation half-life is 6–9 months.

L3

# Trust

The Hallmark.

**L3a Compliance Gates**  
HIPAA, GDPR, SOC 2, EU AI Act

**L3b Quality / Evals**  
Accuracy, hallucination detection, output grading

**L3c Safety & Security**  
Content filtering, adversarial defense, prompt-injection

**L3d Editorial Gates**  
Tone, brand voice, style — the human judgment layer

**L3e Distribution Gates**  
App store approval, ranking, marketplace curation

0 = DEFENSIBLE SUBLAYER (PART OF THE 18-POSITION MOAT MAP)

VERDICT

Essential. More agents = more access control.

HOW PRODUCT LEADERS USE THIS LAYER

- Every regulated buyer maps spend through L3 first. Lead with it in sales.
- L3 is the only layer the model below cannot self-certify (Law IV).
- Editorial taste counts as L3d — yes, it's defensible.

L4

# Access

The Railroad.

**L4a API & Integration**  
REST/GraphQL, SDKs, webhooks

**L4b Agent Protocols**  
MCP, tool-use specs, agent-to-agent comms

**L4c Access Governance**  
RBAC, scoping, audit trails

**L4d Real-Time Infra**  
Streaming, voice, video, low-latency transport

**L4e Agent Identity**  
Verifying which agent acted; credential chains

Ø = DEFENSIBLE SUBLAYER (PART OF THE 18-POSITION MOAT MAP)

VERDICT

Load-bearing walls. Invest accordingly.

HOW PRODUCT LEADERS USE THIS LAYER

- Integrations are weapons. Catalog yours by switching cost, not count.
- MCP and agent protocols are the next 'iOS moment' — pick sides early.
- Agent identity (L4e) will be regulated. Build provenance now.



L5

# Skills

The Jeweler.

## L5a Domain Execution

Legal drafting, code-gen, diagnosis, underwriting

## L5b Decision Scaffolds

Structured thinking patterns, checklists, rubrics

## L5c RAG Workflows

Grounding execution in retrieved context

## L5d Operating Playbooks

Company SOPs, rules, preferences encoded for agents

## L5e Interactional Skills

Tone, empathy, negotiation, the human moment

0 = DEFENSIBLE SUBLAYER (PART OF THE 18-POSITION MOAT MAP)

### VERDICT

Durable if deep. Generic skills get absorbed.

### HOW PRODUCT LEADERS USE THIS LAYER

- This is THE entry point for AI-native startups — pick a deep domain and own it.
- Generic skills get absorbed by L2. Specific playbooks (L5d) don't.
- Codify operating playbooks with your design partners — they become your moat.

L6

# Orchestration

The Store.

**L6a** Agent Loops  
Plan-act-observe cycles

**L6b** Human-in-the-Loop  
Escalation, approval workflows, override design

**L6c** Role Routing  
Decomposition and assignment to right agent

**L6d** Context / State  
Working memory & session state across steps

**L6e** Runtime Assurance  
Monitoring, evals, feedback, drift detection

0 = DEFENSIBLE SUBLAYER (PART OF THE 18-POSITION MOAT MAP)

## VERDICT

Contested. Becoming a feature, not a product.

## HOW PRODUCT LEADERS USE THIS LAYER

- Orchestration is becoming a feature. Don't sell it standalone.
- Human-in-the-loop (L6b) is where buyers actually trust the system. Invest there.
- Build runtime assurance before you scale, not after.

L7

# Expression

The Wearing.

**L7a** Conversational

Voice and chat interfaces

**L7b** Visual / Media

Dashboards, generated images, video, rich media

**L7c** Copilot

Embedded in IDE, email, in-app

**L7d** Transaction

Closes a deal, books, processes payment

**L7e** Async / Ambient

Background agents, nudges, always-on monitoring

Ø = DEFENSIBLE SUBLAYER (PART OF THE 18-POSITION MOAT MAP)

## VERDICT

Modality = commodity. Context = moat.

## HOW PRODUCT LEADERS USE THIS LAYER

- Modality is commodity. Context of use is the moat.
- Embedded (L7c) and transactional (L7d) beat standalone chat over time.
- Ambient surfaces (L7e) are the most underbuilt slot in the stack today.

L8

# Memory

The Ledger.

## L8a Session

Within-conversation context, scratch state

## L8b User Profiles

Persistent preferences, history per user

## L8c Network Learning

Patterns across many customers — fleet intelligence

## L8d Institutional

Docs, decisions, tribal knowledge encoded

## L8e World Models

Accumulated causal understanding of how things work

◻ = DEFENSIBLE SUBLAYER (PART OF THE 18-POSITION MOAT MAP)

### VERDICT

The ultimate moat. Memory that compounds wins.

### HOW PRODUCT LEADERS USE THIS LAYER

- Session memory is table stakes. Network learning (L8c) is the real prize.
- Memory of truth-claims needs an L3 verifier above it (Observation 4).
- Memory compounds — design retention loops on day one, not year three.

# Pick three layers. Defend two. Rent one.

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If you can't draw your product on this stack, you don't yet have a strategy — you have a feature.

The exercise: take any roadmap item and tag it with the layers it touches. If every item is L7, you're a surface — and surfaces get absorbed. If every item is L0/L2, you're competing with trillion-dollar capex. Healthy product strategies span 2–3 layers, with the moat anchored in L1 (data), L5 (skills), L6 (human-in-the-loop) or L8 (memory).

Use the stack in three places this quarter:

- Roadmap reviews — tag every initiative with its layer. Cut anything purely L7 unless context-bound.
- Competitive teardowns — score competitors layer-by-layer. The map shows you what they can't copy.
- Hiring & org design — make sure each defensible layer has an owner. Unowned moats decay.
- Investor narrative — lead with the bottleneck layer (Law II). It's the question they care about.

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